

features or remove existing features) at any point of time, even after purchase. This opens up a lot of scope for further development of the product.

ALPS runs on Linux/Android operating systems. Open source tools, like Eclipse IDE and GNU tools, are used for software development. It is developed using Eclipse integrated development environment (IDE) on Linux platform. The entire software design and source code is available for customisation and users can modify the design to suit more complex projects.

Covering students' needs

One of the biggest challenges faced by developers was covering all peripherals that students typically work on for their projects in a small form factor. The team visited close to 80 engineering colleges for demonstration of their previous product and got to closely work with students to understand their project needs. Based on that experience, the team defined the specification for the carrier card and included interfaces commonly used for components like displays, sensors, communication and I/O.

For real-time applications

Even though there are generic embedded learning platforms available in the market, ALPS is the only product integrated with very high-end technologies specifically designed for the educational segment, claim the developers. Though cheaper, Raspberry Pi sometimes fails to provide high-end video processing required for complex embedded projects.

Another drawback is that Android, a very popular operating system these days, either does not run at all on Raspberry Pi or runs very slow. UDOO is a board that comes at par with ALPS SBC in terms of processing power. But, here the problem is that the interfacing of industry-relevant peripherals can be a difficult task for the student community.

ALPS is a real industrial design and not just a hobby kit. It evidently goes into mass production of real-time

Technical specifications of SBC

Central processing unit (CPU)	- Processor: ARM Cortex A9 i.MX 6 Dual Lite (Quad-compatible) - RAM: 512MB DDR3 SDRAM (double-data rate type-three synchronous dynamic random access memory); expandable up to 2GB
Storage	- On-board Micro SD (secure digital) slot - Standard SD/SDIO (secure digital input/output) slots
Communication interfaces	10/100/1000Mbps Ethernet - Half mini PCIe (peripheral component interconnect express) card connector - Dual USB (universal serial bus) host connector - Micro USB OTG (on-the-go) connector - CAN (controller area network) header - JTAG (joint test action group) header
Audio and video interfaces	- AC97 audio codec with audio out jack and audio in header - HDMI (high-definition multimedia interface) port - LVDS (low-voltage differential signalling) connector with backlight and resistive touch 8-bit CMOS (complementary metal-oxide semiconductor) camera connector - Two lanes MIPI (mobile industry processor interface) camera connector
Input power	5V, 2.5A
Form factor	Pico-ITX (10cm×7.2cm)

projects involving medical electronics, automotive systems, industrial automation, mobile technologies and multimedia.

Taking it to students

ALPS is a finished product and is available in the market. Along with the kit, iWave provides assistance in setting up the advanced embedded system laboratory in those institutions, installing software and drivers, training the faculty and students on how to use the kit and providing technical support for the development of industry-relevant projects. The firm is already associated with more than 50 engineering colleges across the country.

The cost of this kit ranges from ₹ 10,000 to ₹ 25,000 depending on the interfaces required by customers. "We are not thinking about much margin to start with," says Kumar Bhagavath. The focus is on get-

Technical specifications of daughter board

Display interfaces	- RGB LCD: 800×480 pixels screen resolution 16×2 character LCD - Four-digit seven-segment LED - Four general-purpose LEDs
Data converters	- Analogue-to-digital converter (ADC) - Digital-to-analogue converter (DAC)
Communication interfaces	- RS232 • RS485 • I2C header - SPI (serial peripheral interface) header
Sensors	- Accelerometer • Ambient light sensor - Proximity sensor • Temperature sensor - Magnetometer
Motor and switch interfaces	- Dual DC motors/single stepper motor - Six POS DIP (dual in-line package) switch - Four push-button switches
Input power	From SBC
Form factor	10cm×10cm

ting the product familiarised in the educational segment. "Once this is achieved, we will work on our margin, not by increasing the product price but by reducing our manufacturing cost," he adds.

ALPS kit comes with a number of basic and advanced projects and detailed documentation. These, along with the elaborate user guide, are meant to help students get started with the kit without any complications and gain easy hands-on experience with it. ●